



Scalable eco-friendly fabrication of hierarchical superhydrophobic metal surfaces for atmospheric water harvesting

Laylan B. Hassan , Nawzat S. Saadi , Tansel Karabacak

Received: 31 March 2026 / Revised: 13 May 2026 / Accepted: 16 May 2026
© The Author(s) 2026

Abstract The worldwide shortage of freshwater has fueled interest in atmospheric water harvesting (AWH) as a promising option for decentralized water generation. In this study, a scalable method for the fabrication of superhydrophobic (SHP) metal surfaces by sandblasting and hot-water treatment (HWT) followed by fluorination is introduced to improve AWH performance. Aluminum alloy, copper, and zinc substrates were designed to have hierarchical micro-/nanostructures and then fluorinated to render them SHP. Of these, copper showed superior water collection efficiency owing to its high thermal conductivity and optimized surface texture. Under controlled humidity and temperature, water collection on copper was enhanced by 80% compared to aluminum alloy and 34% compared to zinc. To further enhance condensation and droplet drainage, bioinspired groove patterns were fabricated on copper surfaces. When tilted at 45°, the grooved samples exhibited a 21% increase in water collection efficiency compared to flat SHP copper surfaces, owing to improved droplet coalescence and guided drainage. Overall, this study highlights the critical roles of substrate material, hierarchical surface design, and tilt angle in advancing passive atmospheric water-harvesting technologies.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11998-026-01368-9>.

L. B. Hassan (✉), N. S. Saadi, T. Karabacak
School of Physical Sciences, University of Arkansas at Little Rock, South University Avenue, 2801, Little Rock, AR 72204, USA
e-mail: lbhassan@ualr.edu

L. B. Hassan, N. S. Saadi
Department of Physics, University of Duhok (UoD), Duhok 1006 AJ, Iraq

Keywords Superhydrophobic surfaces, Atmospheric water harvesting, Sandblasting, Hot-water treatment, Hierarchical micro-/nanostructures, Groove textures

Introduction

Shortage of water is one of the significant global concerns, especially for arid and semiarid regions. Atmospheric water harvesting (AWH) is an affordable, promising technology to supplement traditional water resources.^{1–4} In AWH, there are two main regimes: fog collection, in which airborne droplets are intercepted by meshes or textured collectors, and condensation-based harvesting, in which water vapor nucleates and grows on a cooled surface below the dew point.^{5–8} These mechanisms require different surface designs and performance criteria; in the current work, we particularly focus on condensation-assisted harvesting^{9–11} on cooled metallic surfaces.

The efficacy of condensation-based AWH largely relies on the wettability and structure of the surface. Many natural organisms have optimized surface morphologies that enable them to efficiently collect water from the air. Cacti^{12,13} and certain desert lizards use hierarchical grooves and superhydrophobic skin structures, respectively, to channel and transport water to required regions of their bodies.^{14–16}

Inspired by such biological systems, researchers have developed engineered surfaces with either homogeneous (uniform surface energy, e.g., uniformly superhydrophobic/superhydrophilic) or heterogeneous (patterned hydrophilic sites on a hydrophobic background, or vice versa) wettability.^{17,18} Homogeneous surfaces, whether superhydrophobic or superhydrophilic, often perform better than patterned ones in long-term dew or moisture harvesting due to fewer contact-line pinning effects.^{10,19–21} Surface modifications, including texturing, chemical treatments, and tilt

angle, also play critical roles in enhancing condensation and drainage dynamics.^{10,20,22–24}

Recent studies have reported laser-structured metallic surfaces^{25–27} and hybrid nanoparticle coatings achieving enhanced condensation; however, such methods remain costly and energy-intensive, limiting scalability for field deployment. Surface patterning strategies are generally divided into morphological (e.g., grooves, ridges) and chemical (e.g., fluorinated coatings or selective plasma treatments) modifications. While lithography, laser etching, and nanoparticle coatings have been used to create such textures,^{21,23,28–32} these methods can be energy-intensive and environmentally unfriendly. Consequently, there is growing interest in scalable, eco-friendly fabrication approaches.

Copper, aluminum alloy, and zinc materials were shortlisted for this study based on thermal conductivity, wettability, and reactivity,³³ all of which affect the efficacy of condensation-based water-harvesting processes.^{9,34} Copper was shortlisted because of its high thermal conductivities (~ 385 W/m·K), making heat transfer quick and promoting faster condensation and droplet formation. Aluminum, a metal that is moderately thermally conductive (~ 205 W/m·K), and zinc, the less thermally conductive (~ 116 W/m·K) of the materials, was tested to determine the efficacy of condensation processes in terms of different thermal transfer rates^{35,36}; besides, these materials, when treated at high temperatures, develop a thick oxide layer, promoting hierarchical roughness. These materials' properties give a basis for testing what factors affect or affect differently the efficacy of a superhydrophobic (SHP) surface engineered for atmospheric harvesting of water.

This study aims to bridge existing knowledge gaps by exploring systematically the effects of surface wettability, structure, and environment on the efficiency of collecting water. Homogeneous wettability surfaces have been produced on aluminum alloy, copper, and zinc materials using sandblasting and hot-water treatment (HWT) processing methods. Moreover, the influence of substrate surface temperature and humidity of the environment on the efficiency of collecting water was also explored. Additionally, the ability of bioinspired textured surfaces to improve the efficiency of harvesting water was also examined, where grooves were added to test their ability to affect droplet coalescence and drainage. Based on the influence of materials, surface wettability, grooved patterns, and tilt angles, this study can provide knowledge that can be applied for creating efficient systems for harvesting atmospheric water. While AWH in arid regions is a significant focus of current research, there remains a critical need for high-efficiency, scalable surfaces for humid coastal regions where atmospheric moisture is abundant but potable water is scarce.

Methodology

Materials

Aluminum alloy (1100H14), copper, and zinc substrates were purchased from Primary Metals (USA) and used as received. Anhydrous ethanol and hexane, as well as a variable-flow peristaltic pump, were obtained from Fisher Scientific. 1H,1H,2H,2H-Perfluorodecyltrichlorosilane (PFDTs, surface energy ≈ 18.1 mN/m) and 1-dodecanethiol (DDT, $\geq 98\%$) were used as low-surface-energy modification agents and were supplied by Sigma-Aldrich (USA). Deionized (DI) water with a resistivity of 18.2 M Ω ·cm was used for HWT and rinsing steps. Sandblasting was performed using aluminum oxide (Al_2O_3) abrasives with a particle size of 60 mesh.

Fabrication of superhydrophobic (SHP) surfaces

Figure 1 illustrates the steps involved in fabricating SHP surfaces on 4×4 cm² coupons of industrial-grade aluminum alloy (1100H14), copper, and zinc using a dual-step roughening process combining sandblasting and HWT, followed by fluorination. Initially, the substrates were sandblasted with a 60-mesh grit size to introduce microscale roughness.³⁷ Subsequently, they were immersed in DI water at 75°C for specific durations: 10 min for aluminum alloy, 24 h for copper, and 1 h for zinc, to grow metal oxide nanostructures through hot-water treatment (HWT)^{37–39} on the microrough substrates and therefore develop hierarchical micro-/nanostructures.

Lastly, to achieve superhydrophobicity, the roughened metal surfaces were chemically functionalized with low-surface-energy agents. The hierarchically textured aluminum alloy and copper substrates were grafted by vapor-phase deposition of a 5% PFDTs solution (1H, 1H, 2H, 2H-perfluorodecyltrichlorosilane). The samples were placed in a tightly sealed container and then heated in an oven at 90°C for 3 h. Afterward, they were dried under ambient conditions for about 12 h. In contrast, the hierarchical zinc substrates were functionalized by dipping in a 10 mM solution of DDT in ethanol. The mixture was gently stirred in a closed beaker at room temperature for 40 min. After modification, the Zn samples were washed with ethanol and then dried overnight under ambient conditions.

Wettability was assessed through contact angle (CA) measurements using a VCA Optima goniometer at room temperature. The sessile drop method was utilized to determine the static contact angles of all surfaces following self-assembled monolayer (SAM) application. Additionally, sliding angles and dynamic contact angles were measured to evaluate SHP behavior and droplet mobility. Surface morphologies were examined using scanning electron microscopy (SEM);

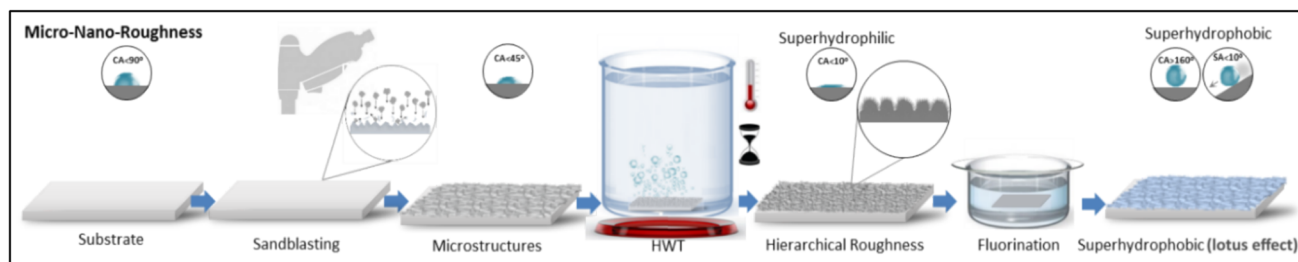


Fig. 1: Schematic illustration of the dual-step roughening process consisting of sandblasting followed by HWT, and the subsequent fluorination step used to achieve low-surface-energy superhydrophobic metal surfaces

JEOL-JSM7000F), while X-ray diffraction (XRD; Bruker D8 Discover) was employed to identify the chemical composition of the oxide layers formed during the HWT process.

Experimental setup for atmospheric water harvesting (AWH)

Figure 2 shows the schematic diagram of the experimental setup of the AWH system. Experiments were conducted under a substrate temperature maintained through a thermoelectric Peltier cooler, keeping the

temperature lower than the dew point. Substrates were placed upright at a 90° angle, and a fog created by a commercial humidifier system was applied, along with a velocity of about 10 cm/s. The laboratory temperature was maintained at 22 °C, and relative humidity (RH) was varied from 95 to 99%. Water droplets condensed on the substrate surface were collected by gravity into a reservoir placed underneath.

The collected quantity of water was measured after a 2 h experimental duration, and each test was carried out thrice to make the results repeatable. For studying the effect of substrate surface properties on the harvesting ability of water, superhydrophobic (SHP)

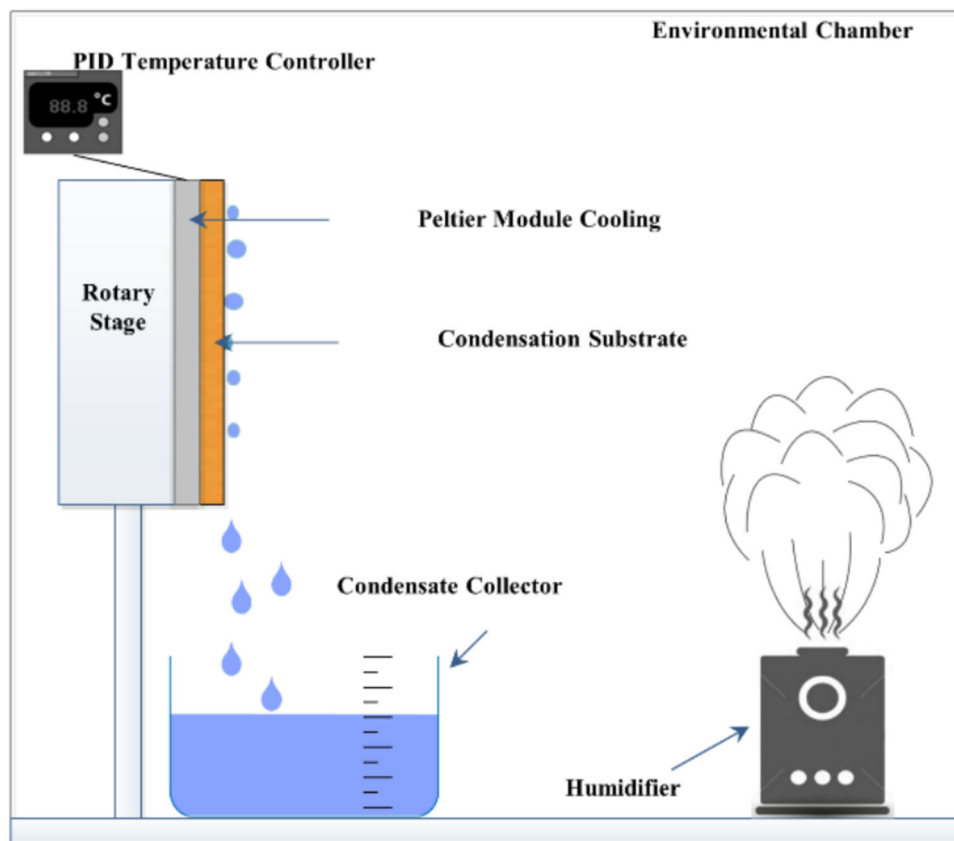


Fig. 2: Experimental setup for AWH. Substrates were mounted vertically on a thermoelectric Peltier cooler and exposed to fog generated by a commercial humidifier. Condensate water is collected and drained by gravity into a container below the substrates

surfaces with micro-/nanoscale roughness underwent different testing conditions. Initially, a test was performed to test the effect of a 90° tilt angle on droplet departure and coalescence rates. Subsequent detailed investigations covered different temperatures of the substrate, such as 5, 10, 15, and 20 °C, along with different RHs of 60, 80, and 99%.

Fabrication of bioinspired grooved copper surfaces

Based on the observed good performance of the copper surface that was previously being used, the design was improved by adding bioinspired grooved patterns taken from the Darkling beetle, *Onymacris unguicularis*.⁴⁰ Grooved copper coupons, each about 3 cm by 3 cm, were created through precision mechanical milling to provide uniform dimensions and positioning for all of the specimens. Each groove had a depth of 4 mm, a pitch of 3 mm, and a groove width of 1 mm. These parameters allowed for the optimization of a capillary-driven drainage path as well as for a controlled water droplet motion created through condensation. Additionally, for the copper specimens, the same methods of creating hierarchical roughness on a planar surface, sandblasting, and hot-water rinsing, were also done for the grooved copper specimens. Subsequent to this, the copper specimens underwent the same PFDS fluorination procedure to provide a copper surface that was now superhydrophobic. The water collection efficiency for the flat, superhydrophobic copper surface, as well as for the grooved, superhydrophobic copper surface, is provided in Fig. 3.

Results and discussion

Superhydrophobic (SHP) hierarchical surfaces: wettability and condensation performance

Developing scalable surface-engineering approaches for AWH remains a significant scientific and societal challenge. In this study, we report a simple method for fabricating hierarchical SHP surfaces on aluminum alloy, zinc, and copper substrates by combining sandblasting and hot-water treatment (HWT) with a subsequent low-temperature fluorination step. Sandblasting and HWT generate micro-/nanoscale hierarchical textures on the metal substrates using only mechanical processing and DI water during the roughness-forming stage, while the final fluorination step provides the low surface energy needed to achieve superhydrophobicity. This dual-step strategy yields surface architectures that are well suited for effective condensation and droplet shedding, analogous to natural water-harvesting surfaces.

Static water contact angle values for untreated control metal surfaces were first measured to establish the wettability baseline prior to modification and were $61.1 \pm 1.2^\circ$, $78.0 \pm 1.4^\circ$, and $27.0 \pm 1.8^\circ$ for aluminum alloy, copper, and zinc, respectively.^{37,41} These measurements indicate that aluminum alloy and copper are moderately hydrophilic, while zinc is highly hydrophilic due to its surface oxide chemistry. After the combined sandblasting and HWT treatment, all samples became highly wettable, consistent with the formation of high-surface-area oxide nanostructures. Subsequent fluorination of the hierarchically textured metals resulted in a pronounced enhancement in water

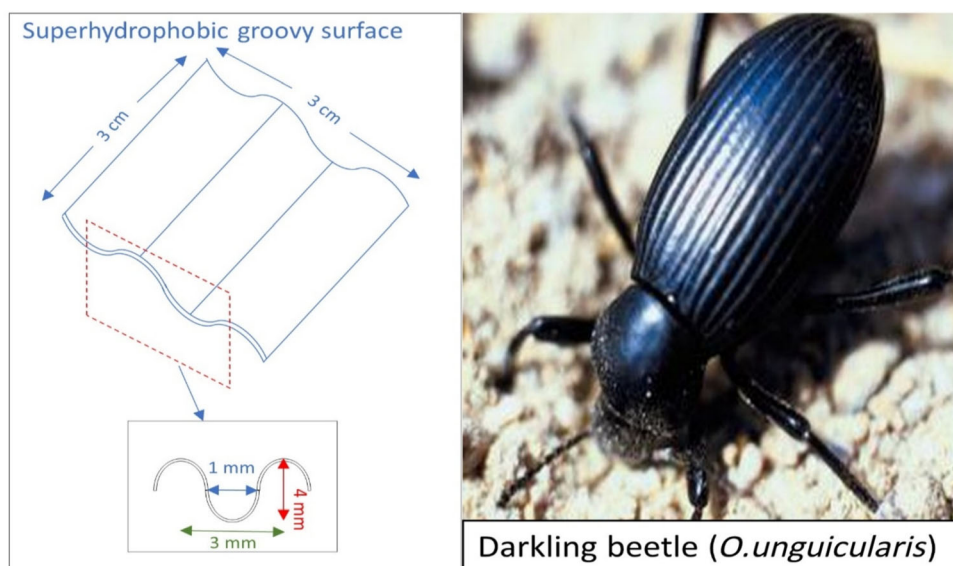


Fig. 3: Schematic of the superhydrophobic grooved copper surface inspired by the Namib Desert Darkling beetle (*Onymacris unguicularis*). Each sample has lateral dimensions of 3 × 3 cm. The groove cross section (inset) shows a width of 1 mm, a pitch (center-to-center distance) of 3 mm, and a height of 4 mm. These dimensions are selected to promote directional droplet transport and efficient drainage during condensation

repellency, with CAs of $161.0 \pm 1.4^\circ$ for aluminum alloy, $164.0 \pm 1.2^\circ$ for copper, and $161.0 \pm 1.2^\circ$ for zinc and sliding angles below 10° . These results clearly confirm the effective transformation of the textured substrates into SHP surfaces capable of sustaining efficient condensation and droplet shedding.

As shown in Fig. 4, scanning electron microscopy (SEM) micrographs reveal dramatic changes from the smooth control surfaces to well-textured surfaces: nano-grass morphologies on aluminum,^{37,38,42} nano-wires on zinc,^{37–39,43} and nano-leaf morphologies on copper.^{37–39,41,43,44} Such nanostructures, formed during HWT, complement sandblasting-generated microscale topographies to achieve dual-scale roughness, which is necessary to maintain the Cassie–Baxter wetting state and enhance capillary-driven droplet mobility. To further elucidate the chemical processes responsible

for the resulting surface nanostructures, XRD characterization was carried out for the HWT-treated aluminum alloy, zinc, and copper substrates. The XRD patterns in Fig. 5 revealed typical diffraction peaks for Al_2O_3 , ZnO , and $\text{CuO/Cu}_2\text{O}$ phases, reflecting successful growth of metal oxide layers on all substrates.^{38,41} These oxide phases are known to provide nanoscale texture under HWT and chemical stability of the surface over long timescales. The presence of such oxide films is consistent with the morphological evidence from scanning electron microscopy (SEM) and confirms that our chemical-free, low-energy HWT process is a feasible means of synthesizing functional oxide nanostructures needed to obtain superhydrophobicity and high condensation performance.

The work proves not only the ability to repeatedly synthesize oxide nanostructures on different metals

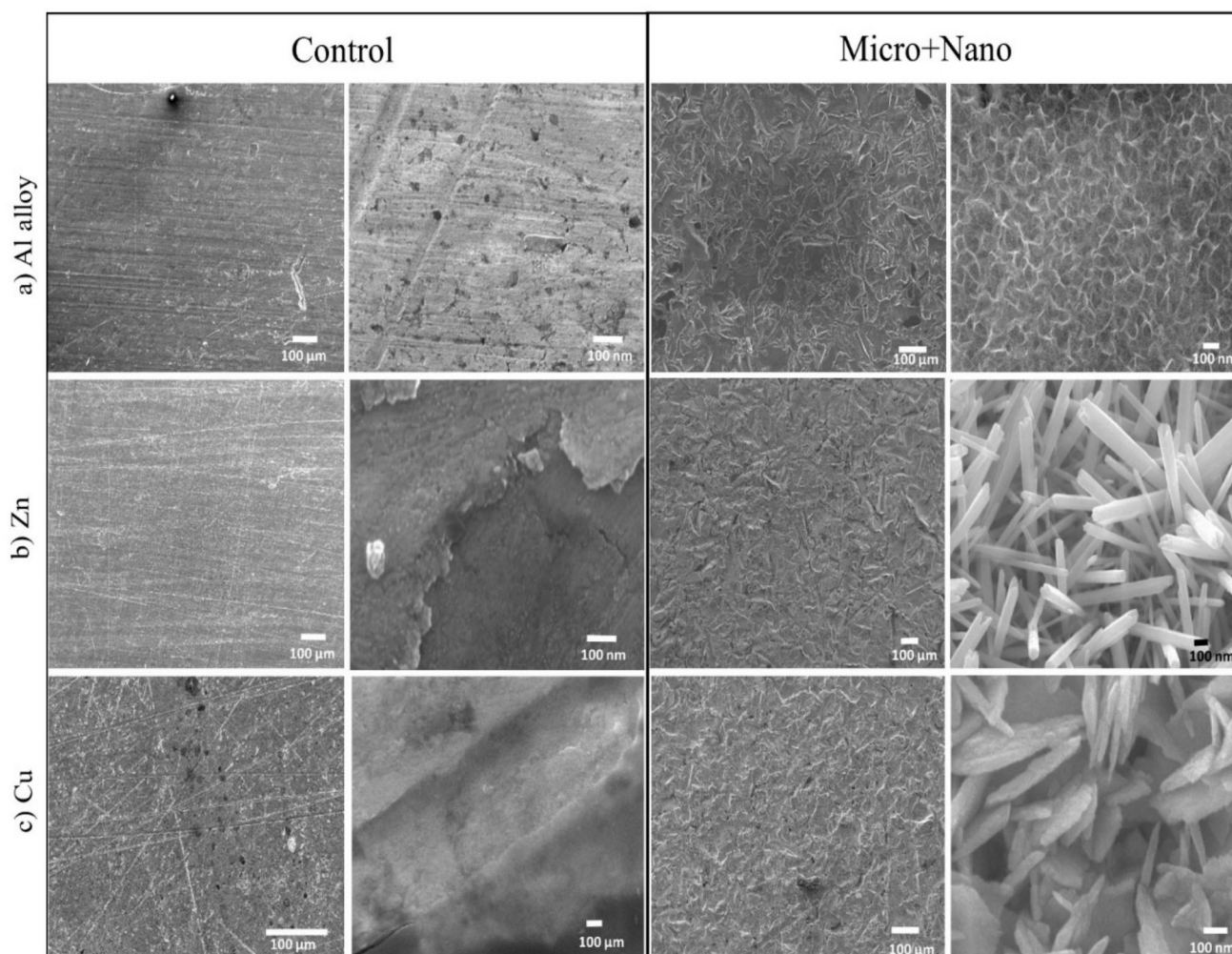
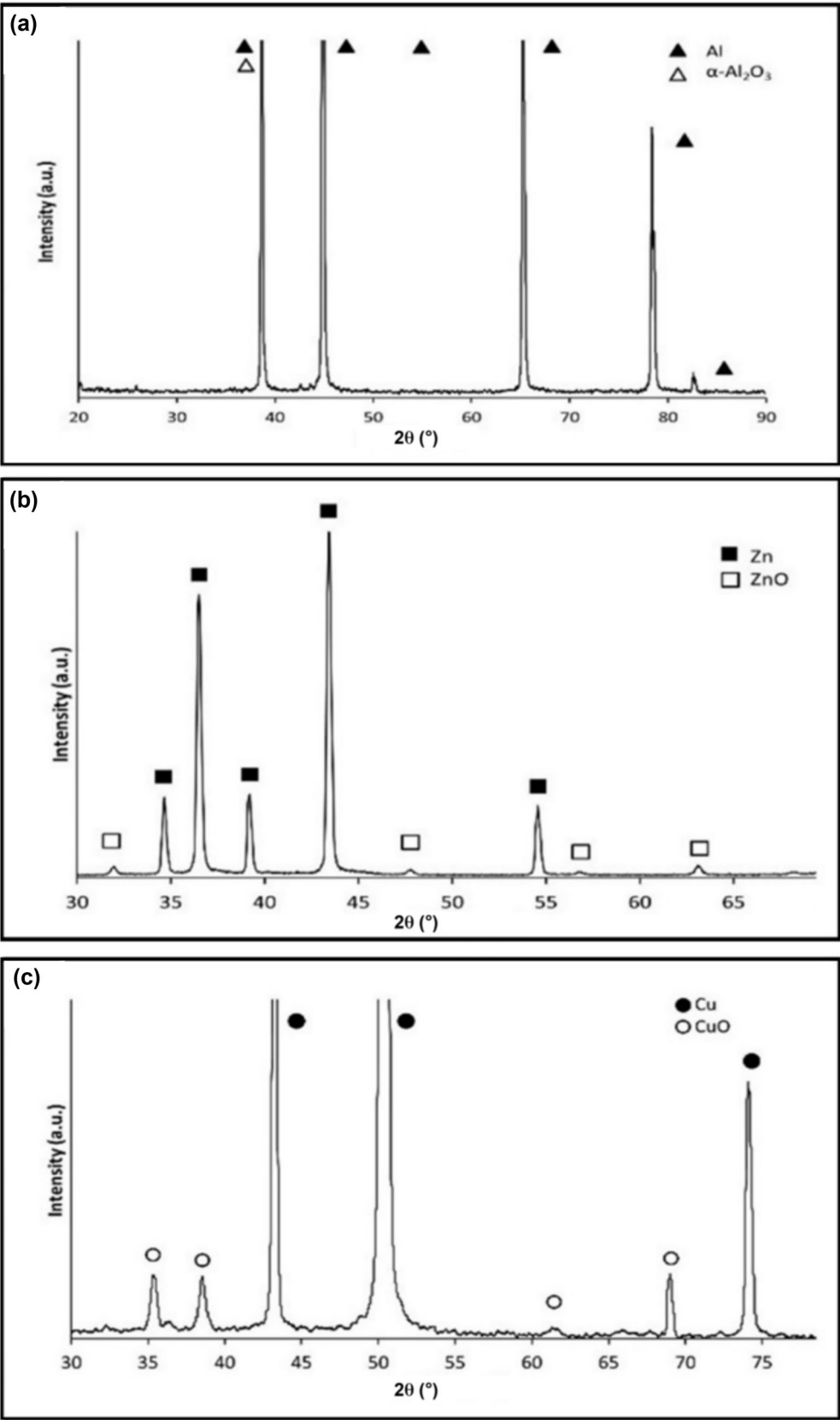


Fig. 4: Scanning electron microscopy (SEM) images comparing the surface morphologies of aluminum alloy (a), zinc (b), and copper (c) substrates in their untreated state (control, left panels) and after combined sandblasting and HWT (micro + nano, right panels). The treated surfaces exhibit well-defined hierarchical micro/nanostructures: nano-grass-like textures on Al, needle-like nanowire formations on Zn, and leaf-like nanostructures on Cu. These dual-scale features, visible in the higher-magnification images (rightmost panels), confirm the successful generation of roughness necessary for superhydrophobicity and efficient AWH



◀Fig. 5: X-ray diffraction (XRD) spectra of the (a) aluminum alloy, (b) zinc, and (c) copper substrates after hot water treatment show distinct diffraction peaks associated with Al_2O_3 , ZnO , and $\text{CuO/Cu}_2\text{O}$ phases. These results confirm the successful formation of oxide nanostructures on the treated surfaces. The presence of these oxide layers provides a stable foundation for the development of hierarchical surface features that contribute to enhanced wettability control and condensation performance

using SB + HWT without resorting to any supplementary additives even for the roughness formation step, but also offers a comparative basis that draws on the differences of thermal conductivity and surface chemistry of the substrates. This is a topic of high relevance that is seldom considered, although its importance is

immense in the realm of atmospheric water harvesting (AWH).

Figure 6 depicts optical micrographs taken in a time-lapse manner of the droplet condensation occurring on the SHP aluminum alloy, zinc and copper surfaces that were subjected to identical test conditions. The in situ images indicate that the droplet condensation characteristics and behavior of droplets on the copper surface are superior compared to the droplets condensing on either the zinc or aluminum alloy surfaces, as demonstrated by the higher rate of nucleation and the more rapid movement of the droplets on the copper.⁹ The data collected in this study support the conclusion that the substrate material has a significant impact on the efficiency of the water-harvesting process and therefore support the continuation of optimizing the copper substrate. Conversely, sandblasted and HWT-treated surfaces that were not fluorinated exhibit superhy-

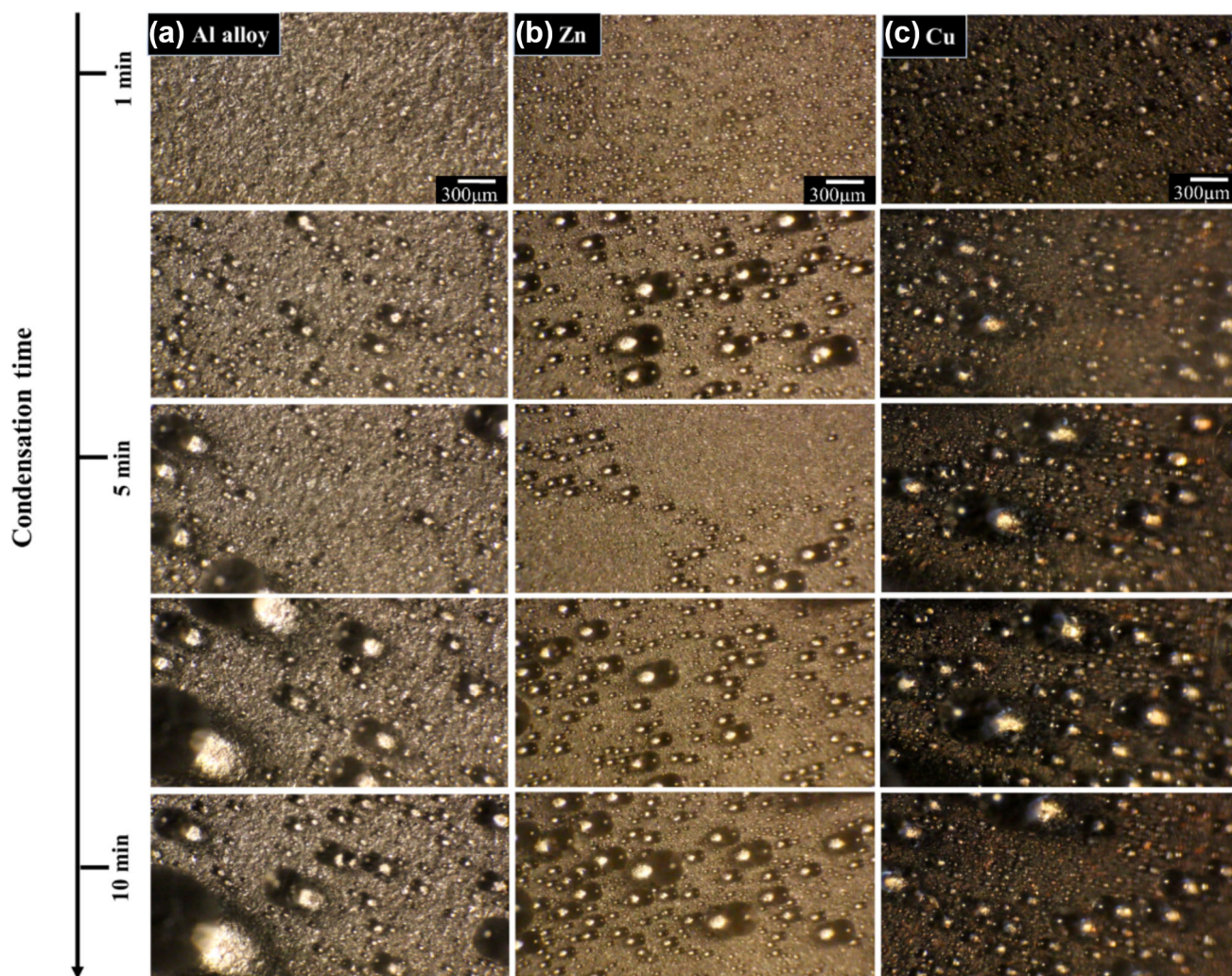


Fig. 6: Time-lapse optical microscopy images showing water droplet condensation on (a) aluminum alloy, (b) zinc, and (c) copper SHP surfaces at a horizontal orientation (0° tilt). Images were recorded at 1, 5, and 10 min. Copper demonstrates faster droplet nucleation and higher mobility, supporting its superior water-harvesting performance compared to Zn and Al alloy

drophilicity,³⁷ as evidenced by the immediate formation of a continuous water-film layer that covers the entire surface of the substrate during droplet condensation. The water film presents a strong barrier to the dropping of the droplets and renders the removal of the water from the substrate virtually ineffective; therefore, these unmodified sandblasted and HWT-treated substrates are not suitable for effective water collection.

While other presently available methods depend on expensive technology such as sophisticated lithography²⁸ or expensive laser patterning,²¹ our low-cost fabrication method offers a novel, scalable, and environmentally responsible pathway to develop practical AWH surfaces. This work fills a critical gap in the literature by demonstrating that simple physical treatments can yield high-performance water-repellent surfaces without compromising effectiveness or sustainability.

Water collection performance of SHP aluminum alloy, copper, and zinc surfaces under varying environmental conditions

SHP surfaces made of aluminum alloy, copper, and zinc with water contact angles over 150° and sliding angles under 10° were tested for their ability to collect water from the air.³⁷ As shown in Fig. 7, the SHP copper surface collected the most water, about 11.2 mL, followed by SHP zinc with 7.35 mL, and SHP aluminum alloy with 2.24 mL. In comparison, the untreated control surfaces collected much less: copper 1.6 mL, aluminum alloy 0.8 mL, and zinc 0.56 mL. This shows that SHP surfaces are much more effective than untreated ones, mainly because they promote dropwise

condensation, which helps water droplets roll off easily unlike filmwise condensation, where water sticks to the surface.⁹

Out of all the SHP materials tested, copper showed the most impressive results. This can be attributed largely to its excellent thermal conductivity, allowing it to dissipate heat rapidly and facilitate condensation more effectively.⁵ Although all samples underwent the same surface treatment and coating, each type of metal developed distinct oxide nanostructures, resulting in varying surface textures. Given that the testing environment was consistent, the performance differences are primarily linked to the inherent properties of the materials, with copper's high thermal conductivity being a significant factor.

To see how temperature impacts water harvesting, we ran experiments at 5, 10, 15, and 20°C , all with 99% humidity, as shown in Fig. 8a. At 5°C , ~ 20 mL was collected by copper, ~ 14.24 mL by zinc, and ~ 15 mL by aluminum alloy. Moving to 10°C , copper was still the top performer, collecting about 15 mL, followed by zinc with about 12.5 mL and aluminum alloy with around 10 mL. At 15°C , copper collected around 12.8 mL, zinc about 10 mL, and aluminum alloy dropped to about 5.6 mL. To end with 20°C , copper accumulated about 11 mL, while zinc and aluminum alloy collected approximately 7.4 and 2.24 mL, respectively. These findings suggest that as temperature rises, the total water collected tends to decrease, though copper consistently outperforms zinc and aluminum alloy.

The analysis of percentage difference shows that for a temperature of 5°C , copper collected about 22.4% more water than the aluminum alloy, and about 28.8% more than zinc. For a temperature of 10°C , copper collected about 29.7% more water than aluminum

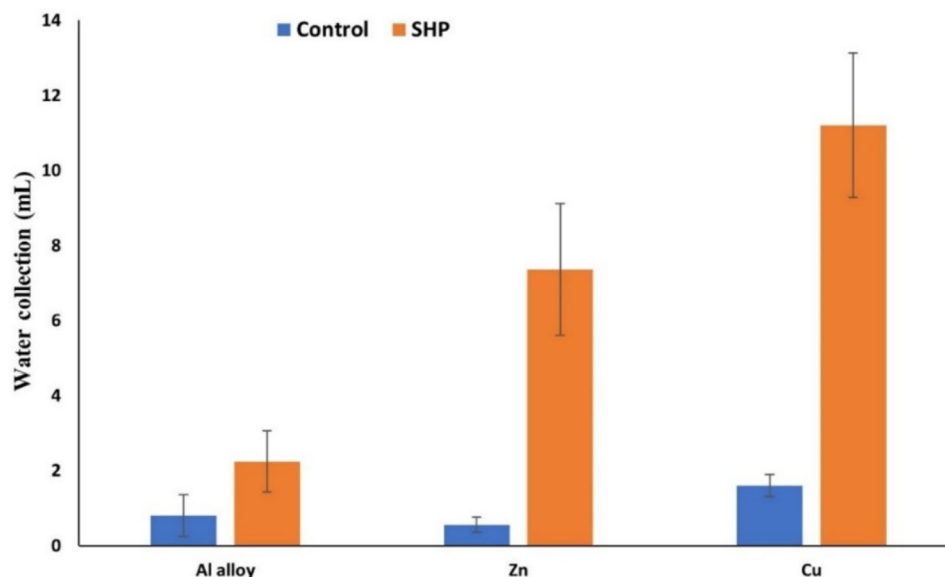


Fig. 7: Water collection comparison between control and SHP surfaces on aluminum alloy, zinc, and copper. SHP surfaces show significantly higher collection, with copper achieving the highest yield. Error bars indicate standard deviation

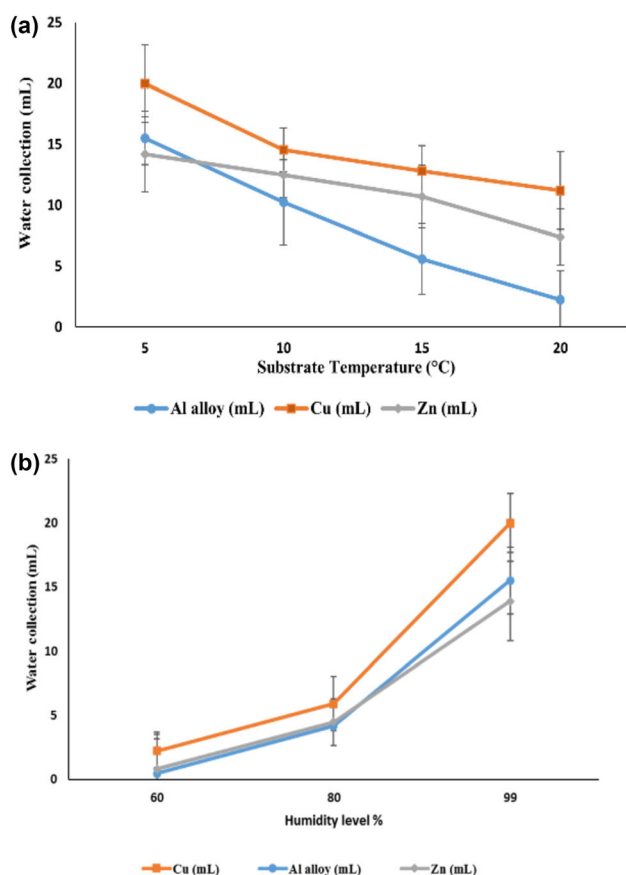


Fig. 8: (a) Effect of substrate temperature on water collection efficiency for SHP aluminum alloy, copper, and zinc surfaces. Water collection decreased with increasing substrate temperature. (b) Influence of RH levels (60, 80, and 99%) on water harvesting. Higher humidity levels resulted in greater water collection

alloy, and about 14.3% more than zinc. For a temperature of 15 °C, the difference was even larger, such that copper collected about 56.3% more water than aluminum alloy, and about 16.3% more than zinc. For a temperature of 20 °C, copper was even better, whereby it collected 80% more water than aluminum alloy, and approximately 34.3% more than zinc. It is, therefore, apparent that thermal conductivity is a strong factor that affects the efficiency of the condensed collection of water from the air.

Relative humidity (RH) was also found to be a key factor that affects the yield of condensation, as demonstrated in Fig. 8b. Experiments performed under 60, 80, and 99% RH conditions made it clear that the increase in humidity resulted in higher water accumulation on all materials. Under 60% RH, copper accumulated approximately 2.24 mL of water, while aluminum alloy and zinc accumulated about 0.48 mL and 0.8 mL, respectively. Under 80% RH, copper accumulated around 6 mL of water, unlike approximately 4.16 mL for aluminum alloy and about 4.48 mL for zinc. Under nearly saturated conditions of 99%

RH, copper accumulated nearly 20 mL, compared to about 15 mL for aluminum alloy and about 14 mL for zinc. Under all humidity conditions, copper performed better than the other materials, because of its high thermal conductivity property, making it easier to retain optimal surface temperature and, therefore, facilitate continuous condensation. For example, under 60% RH, copper condensed about 78.6% more water than aluminum alloy and about 64.3% more than zinc. Under 80% RH, copper accumulated about 29.8% more water than aluminum alloy and about 24.3% more than zinc. Under 99% RH, copper captured about 22.4% more water than aluminum alloy and about 30.4% more than zinc.

Overall, these results show that both the thermal properties of the substrate and the surface treatment (such as hydrophobic coatings) are important for improving condensation efficiency. Here, condensation efficiency is defined as the ratio between the actual amount of condensed water collected on a surface and the theoretical maximum amount of water vapor available. Copper's ability to maintain a good surface temperature and to promote droplet removal increases its efficiency in passive water-harvesting systems under different humidity conditions.

Performance enhancement of AWH using grooved SHP copper surfaces with optimized tilt angles

Previous AWH studies were predominantly conducted on planar SHP surfaces.^{40,45–47} In nature, however, many organisms possess rough surface textures that enhance water harvesting, such as the exoskeletons of desert beetles.^{16,20,21,23,40,45} The effect of surface texture was studied here by synthesizing SHP copper surfaces with a grooved pattern to mimic such natural architectures and comparing these with planar SHP copper surfaces. In addition, the effect of surface orientation was investigated by testing flat and grooved surfaces at 90° and 45° tilt angles.

At a vertical position of 90°, as shown in Fig. 9, the grooved SHP copper surface captured significantly more water than the flat one under identical conditions. This is made possible by the groove structure, which encourages the speedy movement of droplets. Condensed droplets can easily separate from the ridges and move toward the valleys between the grooves because of the pressure exerted by the curvatures. This process enables droplet coalescing and hence the promotion of droplet accumulation. When droplets coalesce and depart, they also free up the surface for new nucleation sites, so continuous and effective condensation can occur.

Sample tilt angles were also investigated to compare their effect on water collection efficiency. Figure 9 reveals that the water collected on a flat SHP copper surface with a tilt angle of 45° was nearly twice that for the same surface mounted vertically at 90°. A similar trend was also found for the grooved SHP copper

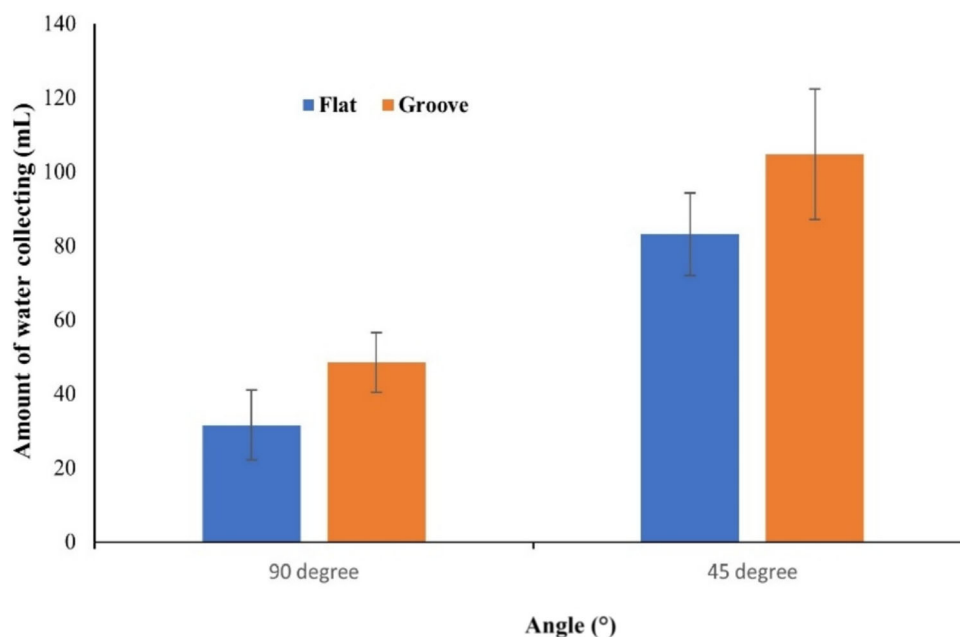


Fig. 9: Comparison of water collection rates for grooved and flat SHP copper surfaces at two mounting angles (90° and 45°). Grooved surfaces at 45° tilt showed the highest water collection efficiency

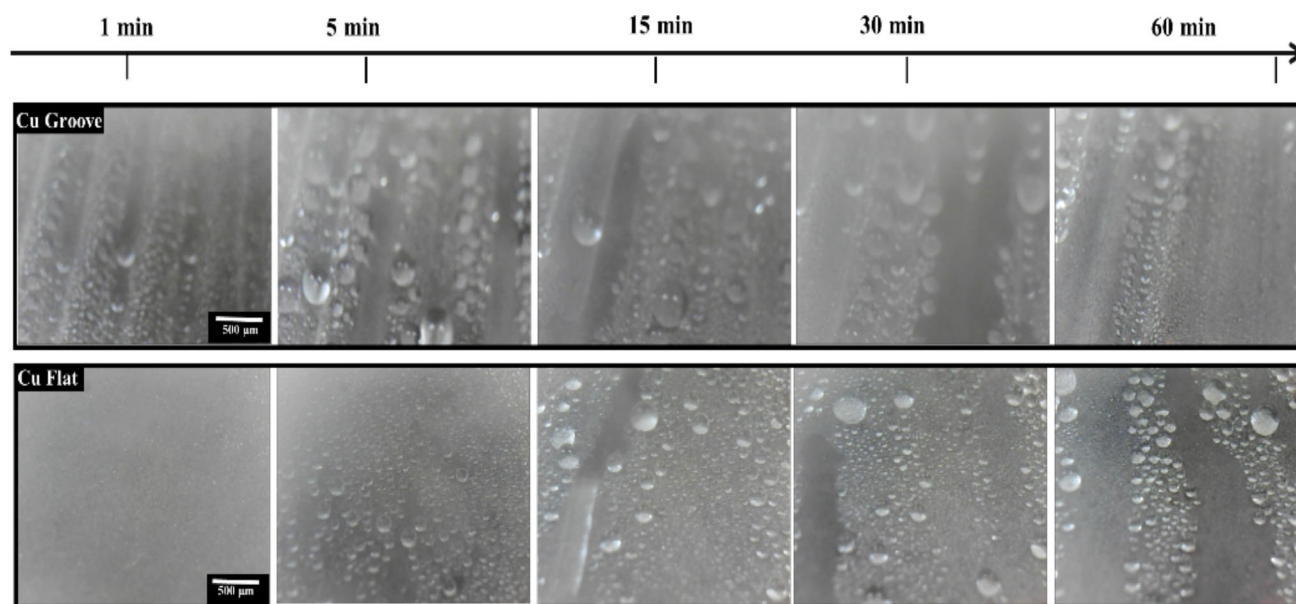


Fig. 10: Time-lapse condensation on grooved (top) and plane (bottom) copper surfaces tilted at 45° for 60 min. The grooved surface exhibits enhanced droplet drainage and coalescence, allowing for sustained condensation and improved water collection efficiency

surface; the 45° tilt angle showed significantly better water collection compared to the vertically oriented grooved sample. Such a large enhancement probably happens because an intermediate tilt provides the best compromise between gravitational and available condensation area.

Figure 10 shows a comparison of time-lapse condensation between grooved and flat copper surfaces

inclined at 45° for 60 min. The grooved copper surface (top row) was able to retain condensed droplets, while simultaneously promoting guided drainage, coalescence, and rapid detachment. Compared to other surfaces, the relatively lower mobility of droplets reduces the mobility of the droplets and accentuates stagnation on the plane copper substrate (bottom row), thus hindering the efficiency of water harvesting.

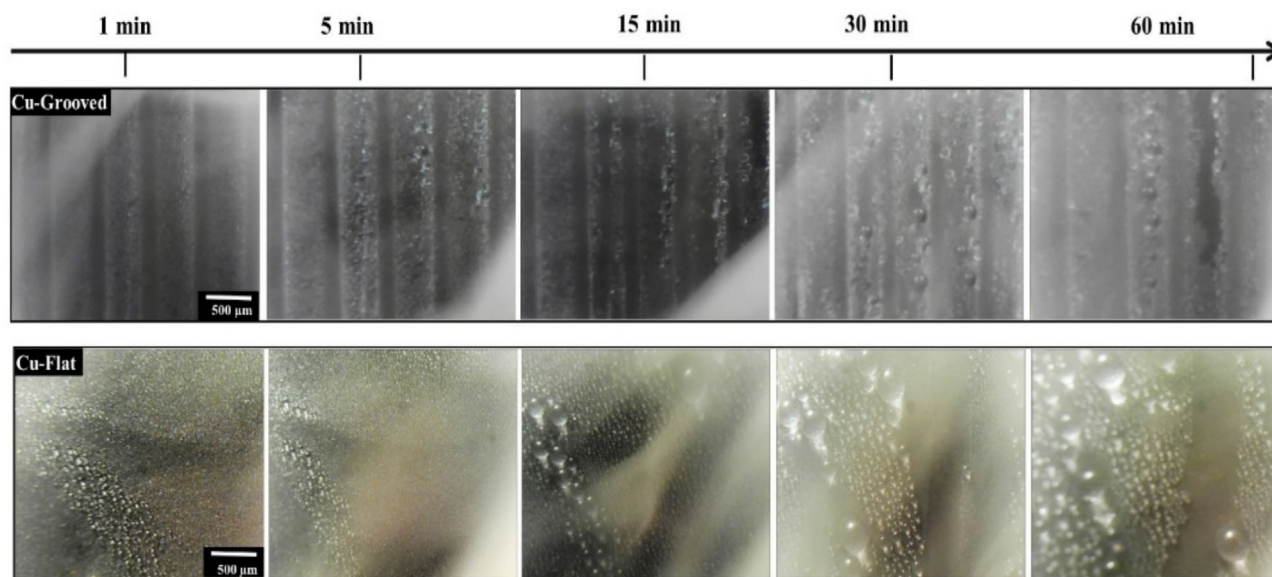


Fig. 11: Time-lapse condensation on grooved and flat SHP copper surfaces with a tilt angle of 90° over 60 min. The copper groove surface (top) promotes rapid gravitational drainage, leading to faster droplet removal but less condensation and coalescence. The copper flat surface (bottom) shows slower droplet movement and higher retention

Through optimization of the surface topography relative to the gravitational forces, the designed surface topographies facilitate mobility of droplets and optimize the condensation processes, thus improving their efficiency. These results thus illustrate grooved surfaces as an effective design strategy toward environmentally friendly atmospheric water collection devices. See the supplemental videos S1 and S2 for more information.

When time-lapse images of condensation processes on flat and grooved copper substrates at a 90° tilt over 60 min are examined (Fig. 11), condensate droplets on the grooved surface are seen to move rapidly downward under gravity, efficiently clearing the surface. However, the vertical orientation of the grooves not only causes droplets to fall quickly but also reduces the time available for condensation and hampers droplet merging, leading to smaller droplets and less stored water. In contrast, flat copper surfaces allow for higher accumulation of droplets and much slower droplet motion, resulting in stagnation. Although grooves can channel liquid, completely vertical grooves severely restrict the area effectively used for condensation, thus reducing the total amount of water collected.

It is observed that for a 45-degree groove, there is a greater efficiency of atmospheric water harvesting than for a horizontal, as well as a 90-degree, surface. Though the faster gravity-driven removal of water at a 90-degree angle leads to efficient drying of the surface, at the same time, this decreases the rate of condensation. The tilted surface of 45 degrees is more efficient for harvesting atmospheric water by 21 percent than a horizontal surface. This is because of efficient droplet motion, merging, and lesser accumu-

lation of droplets. The horizontal surface of 0 degrees, being fully open to the atmospheric air, is non-effective for droplet motion, favoring film condensation.

The onset of droplet motion on textured SHP surfaces at inclined angles can be predicted by balancing the gravitational driving force with contact-line pinning and inertial resistance.^{48–51} The gravitational force depends primarily on the liquid density, droplet volume, and contact diameter, while the pinning force is governed by the surface tension and the difference between the advancing and receding contact angles. At higher tilt angles, the detachment criterion is more easily satisfied, leading to earlier droplet release, shorter coalescence times, and reduced travel distances. At intermediate tilt angles, droplets grow to larger sizes before sliding off, acting as mobile collectors that sweep over larger areas and transport smaller droplets along the grooves, thereby prolonging drainage time.⁵¹ At tilt angles below 20°, however, the driving force is insufficient to overcome pinning, which limits the area that can be cleared.

Premature detachment of sub-optimal, small droplets removes them before they have acquired sufficient volume and momentum to trigger cascades of coalescence and sweeping with broad tracks. The cleared area in each shedding event is therefore small, and the surface is quickly re-nucleated with small droplets and minimal dry-zone formation, which reduces the time-averaged collection rate compared to the larger, sliding collector droplets at 45°.

In summary, this research verifies that surface patterning and tilt angle are key parameters controlling droplet behavior and water collection efficiency. Of the designs tested, grooved copper surfaces at 45° maxi-

Table 1: Summary of recent AWH surfaces, with fabrication method, water collection performance, and sustainability features

Surface Type / Method	Water collection performance (L/m ² -h)	Sustainability aspects	References
Sandblasting + HWT on Cu/Al/Zn with SHP coating	SHP Cu: ~ 5, 21% gain via grooves & 45° tilt	Scalable, low cost, eco-friendly	This work
Laser micropatterned aluminum (no coating)	~ 0.3	No coating needed, robust, emissive cooling, energy-intensive laser	21
Nature-inspired micro-/nanostructuring aluminum + wettability gradient	~ 0.85	Chemical etching and laser, less scalable	31
Patterned cellulose film (SHP/SHPH) + Ag/TiO ₂	~ 10.43	Low cost, antibacterial, easy fabrication	23
Patterned hybrid SHP/SHPH polymer surface	Estimated up to ~ 5.2 based on 78% improvement over SHP and 536% over SHB surfaces; actual value not explicitly reported.	Polymer-based, some chemicals used	32
Electrospinning blended cellulose acetate/polystyrene fibers	~ 0.62 at 80% RH	Cost-effective, scalable, low energy	52
Radiative cooling via cellulose-based fabric, Hierarchical cellulose fabric	~ 0.04 at 80% RH	Low cost, passive cooling, large-scale deployment	53

mized condensation, coalescence, and drainage and are therefore optimal candidates for passive large-scale atmospheric water collection in drylands. Supporting videos S3 and S4 further visualize dynamic condensation behavior on such optimized surfaces.

Based on this discovery, we hereby report a novel, scalable and sustainable fabrication method to fabricate hierarchical SHP metal surfaces by simple mechanical sandblasting and subsequent HWT. This top-down method presents a favorable alternative to traditional AWH surface modifications based on sophisticated lithography, chemical etching, or high-energy laser treatment. Unlike them, however, our method attains excellent superhydrophobicity without involving any toxic chemicals or high-tech instrumentation, making it particularly convenient for use in off-grid and resource-limited applications.

As highlighted in Table 1, the existing method of manufacture is a marked improvement over previous agricultural water-harvesting methods, namely because of its ease of manufacture, scalability, and biocompatibility. For example, whereas most existing methods require the use of a laser micromachining tool, or require polymers doped through a chemical process, or require the aid of electro-spinning, the existing method creates these nanostructures through purely physical processes for a host of metals such as copper, aluminum, or zinc.

Our design takes clear inspiration from nature's water-harvesting strategies in, for example, cactus spines and Namib Desert beetles. By emulating such biologically optimized textures and drainage routes—

most prominently in grooved copper structures—we will facilitate directional droplet transport and effective surface clearing. In this way, we will optimize performance without sacrificing integrity to sustainable, minimalist design principles via a bioinspired engineering strategy and, in so doing, help bridge the gap between bionic concepts and practical applications.

Conclusions

This research introduces a scalable approach to the production of superhydrophobic metal surfaces for improving AWH. By combining sandblasting with hot-water treatment (HWT), hierarchical micro-/nanostructures were fabricated on aluminum alloy, zinc, and copper substrates, and subsequent fluorination rendered these textured surfaces superhydrophobic. Among the investigated metals, copper surfaces exhibited the highest water collection efficiency, which is attributed to their high thermal conductivity and favorable surface morphology. Under controlled humidity and temperature conditions, water collection on copper increased by approximately 80% compared to aluminum alloy and by 34% compared to zinc. The addition of bioinspired grooved textures on copper further promoted droplet coalescence and mobility, leading to enhanced water harvesting, particularly at a tilt angle of 45°, where a ~ 21% increase in water collection was achieved relative to flat SHP copper surfaces.

The overall fabrication route relies on simple mechanical and thermal treatments followed by a single low-temperature surface modification step and can be applied to common metal substrates over large areas. The integration of precision surface engineering and bioinspired design therefore presents a promising direction for the development of scalable, passive AWH systems that contribute to addressing global water scarcity.

Acknowledgments The authors thank the UALR Center for Integrative Nanotechnology Sciences for helping with SEM imaging.

Authors' contributions Mrs. Hassan conceived and designed the experiments. Saadi performed the experiments. Karabacak supervised the project and provided valuable feedback. All authors contributed to writing and editing the manuscript.

Funding This work was supported by NASA (grant no: NNX09AW22A) and NSF (grant nos: EPS-1003970 and 1159830).

Data availability The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Code availability No custom code or software was used in this study. All analyses were performed using standard software and methodologies, as detailed in the manuscript

Conflict of interest The authors declare no conflicts of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Alayli, Y, Hadji, NE, Leblond, J, "A New Process for the Extraction of Water from Air." *Desalination*, **67** 227–229 (1987)
- Bar, E, "Extraction of Water from Air—An Alternative Solution for Water Supply." *Desalination*, **165** 335 (2004)
- Hunter, P, "Turning Nature's Inspiration into a Production Line." *EMBO reports*, **15** (11) 1123–1127 (2014)
- Sultan, A, "Absorption/Regeneration Non-conventional System for Water Extraction from Atmospheric Air." *Renewable Energy*, **29** (9) 1515–1535 (2004)
- Du, C, Huang, C, "A Floating Vapor Condensation Structure in a Heat-Localized Solar Evaporation System for Facile Solar Desalination." *Applied Thermal Engineering*, **201** 117834 (2022)
- Nioras, D, et al. "How Different are Fog Collection and Dew Water Harvesting on Surfaces with Different Wetting Behaviors?" *ACS Applied Materials & Interfaces*, **13** (40) 48322–48332 (2021)
- Wang, F, et al. "A High-Performing Single-Stage Invert-Structured Solar Water Purifier Through Enhanced Absorption and Condensation." *Joule*, **5** (6) 1602–1612 (2021)
- Wang, J, et al. "Atmospheric Water Harvesting: Critical Metrics and Challenges." *Energy & Environmental Science*, **15** (12) 4867–4871 (2022)
- Hassan, LB, Saadi, NS, Karabacak, T, "Enhanced Dropwise Condensation Using Superhydrophobic Surfaces with Hierarchical Roughness: A Scalable and Eco-Friendly Approach." *MRS Bulletin*, **50** 819–829 (2025)
- Lee, A, et al. "Water Harvest via Dewing." *Langmuir*, **28** (27) 10183–10191 (2012)
- Miljkovic, N, Enright, R, Wang, EN, "Effect of Droplet Morphology on Growth Dynamics and Heat Transfer During Condensation on Superhydrophobic Nanostructured Surfaces." *ACS Nano*, **6** (2) 1776–1785 (2012)
- Gurera, D, Bhushan, B, "Passive Water Harvesting by Desert Plants and Animals: Lessons from Nature." *Philosophical Transactions of the Royal Society A*, **378** (2167) 20190444 (2020)
- He, G, Zhang, C, Dong, Z, "Survival in Desert: Extreme Water Adaptations and Bioinspired Structural Designs." *iScience*, **26** (1) (2023)
- Aslan, D, Selçuk, SA, "A Biomimetic Approach to Rain-water Harvesting Strategies Through the Use of Buildings." *Eurasian Journal of Civil Engineering and Architecture*, **2** (1) 27–39 (2018)
- Sun, H, et al. "Bioinspired Micro-and Nanostructures Used for Fog Harvesting." *Applied Physics A*, **127** (6) 461 (2021)
- Zhang, Y, et al. "Lizard-Skin-Inspired Nanofibrous Capillary Network Combined with a Slippery Surface for Efficient Fog Collection." *ACS Applied Materials & Interfaces*, **13** (30) 36587–36594 (2021)
- Chi, D-T, Nguyen, T-B, "Enhancing Water Condensation on Hybrid Surfaces by Optimizing Wettability Contrast." *Surfaces*, **7** (3) 508–516 (2024)
- Cha, H, et al. "Dropwise Condensation on Solid Hydrophilic Surfaces." *Science Advances*, **6** (2) eaax0746 (2020)
- Bhushan, B, Jung, YC, "Natural and Biomimetic Artificial Surfaces for Superhydrophobicity, Self-cleaning, Low Adhesion, and Drag Reduction." *Progress in Materials Science*, **56** (1) 1–108 (2011)
- Parker, AR, Lawrence, CR, "Water Capture by a Desert Beetle." *Nature*, **414** (6859) 33–34 (2001)

21. Pou-Álvarez, P, et al. “Efficient Autonomous Dew Water Harvesting by Laser Micropatterning: Superhydrophilic and High Emissivity Robust Grooved Metallic Surfaces Enabling Filmwise Condensation and Radiative Cooling.” *Advanced Materials*, **37** (18) 2419472 (2025)
22. Narhe, RD, González-Viñas, W, Beysens, DA, “Water Condensation on Zinc Surfaces Treated by Chemical Bath Deposition.” *Applied Surface Science*, **256** (16) 4930–4933 (2010)
23. Xu, C, et al. “Desert Beetle-Inspired Superhydrophilic/Superhydrophobic Patterned Cellulose Film with Efficient Water Collection and Antibacterial Performance.” *ACS Sustainable Chemistry & Engineering*, **6** (11) 14679–14684 (2018)
24. Zhai, L, et al. “Patterned Superhydrophobic Surfaces: Toward a Synthetic Mimic of the Namib Desert Beetle.” *Nano Letters*, **6** (6) 1213–1217 (2006)
25. Lou, D, et al. “Laser Textured Superhydrophobic SiC Ceramic Surface and the Performance of Condensation Heat Transfer.” *Crystals*, **13** (5) 840 (2023)
26. He, J-G, et al. “Fabrication of Metallic Superhydrophobic Surfaces with Tunable Condensate Self-Removal Capability and Excellent Anti-frosting Performance.” *Nanomaterials*, **12** (20) 3655 (2022)
27. Balamurugan, KG, “Investigation on the Effect of Laser Surface Texturing on the Wettability of Pure Zinc Substrate.” *Journal of Engineering and Applied Science*, **71** (1) 142 (2024)
28. Estephan, J, et al. “Controlled Wettability of Biphasic Patterned Surfaces for Enhanced Atmospheric Water Harvesting.” *Micro and Nano Engineering*, **23** 100255 (2024)
29. Garrod, RP, et al. “Mimicking a Stenocara Beetle’s Back for Microcondensation Using Plasmachemical Patterned Superhydrophobic–Superhydrophilic Surfaces.” *Langmuir*, **23** (2) 689–693 (2007)
30. Guo, H, et al. “Super Moisture-Sorbent Sponge for Sustainable Atmospheric Water Harvesting and Power Generation.” *Advanced Materials*, **36** (52) 2414285 (2024)
31. Li, Z, et al. “Nature-Inspired Surface Engineering for Efficient Atmospheric Water Harvesting.” *ACS Sustainable Chemistry & Engineering*, **11** (30) 11019–11031 (2023)
32. Wei, L, Soo, HS, Chen, Z, “Patterned Hybrid Surfaces for Efficient Dew Harvesting.” *ACS Applied Materials & Interfaces*, **16** (38) 51715–51726 (2024)
33. Williams, JJ, “Surface Reactions of Zinc Vapour with Steel Relevant to the Zn-55% Al-1.5% Si Hot Dip Metal Coating Process.” PhD Thesis, University of Wollongong (2005)
34. Lei, C, et al. “Chemistries and Materials for Atmospheric Water Harvesting.” *Chemical Society Reviews*, **53** (14) 7328–7362 (2024)
35. Kuznetsov, YI, Semiletov, A, “Development of Methods for Inhibiting the Corrosion of Metals and New Options for Their Application: A Review. Part II Hydrophobization of Metal Surface as a Way to Inhibit Corrosion.” *International Journal of Corrosion and Scale Inhibition*, **13** (4) 2515–2542 (2024)
36. Zhang, E, “Investigation into the Condensation Effect on the Greenhouse Operational Energy Performance and Energy Saving Strategies.” PhD Thesis, The Pennsylvania State University (2025)
37. Hassan, LB, Saadi, NS, Karabacak, T, “Hierarchically Rough Superhydrophobic Metal Surfaces Fabricated by a Sandblasting and Hot Water Treatment Process.” *The International Journal of Advanced Manufacturing Technology*, **137** 4051–4061 (2025)
38. Saadi, NS, Hassan, LB, Karabacak, T, “Metal Oxide Nanostructures by a Simple Hot Water Treatment.” *Scientific Reports*, **7** (1) 7158 (2017)
39. Saadi, NS, Hassan, LB, Karabacak, T, “Scalable and Durable Superhydrophobic-Superoleophilic Nanostructured Zinc-Coated Steel Wool for Efficient Oil-Water Separation.” *Journal of Environmental Chemical Engineering*, 119644 (2025)
40. Nørgaard, T, Dacke, M, “Fog-Basking Behaviour and Water Collection Efficiency in Namib Desert Darkling Beetles.” *Frontiers in Zoology*, **7** (1) 23 (2010)
41. Hassan, LB, Saadi, NS, Karabacak, T, “Hierarchically Rough Superhydrophobic Copper Sheets Fabricated by a Sandblasting and Hot Water Treatment Process.” *The International Journal of Advanced Manufacturing Technology*, **93** 1107–1114 (2017)
42. Hassan, LB, Saadi, NS, Karabacak, T, “Fabrication of Robust and Durable Superamphiphobic Aluminum Alloy and Zinc Surfaces via Dual Sandblasting and Steam Treatment.” *The International Journal of Advanced Manufacturing Technology* (2025)
43. Saadi, NS, et al. “Growth of Zinc Oxide Nanowires by a Hot Water Deposition Method.” *Nanotechnology*, **36** (3) 035601 (2024)
44. Saadi, NS, et al. “Special Wettable Nanostructured Copper Mesh Achieved by a Facile Hot Water Treatment Process.” *Materials Research Express*, **4** (9) 095021 (2017)
45. Yan, X, et al. “Atmosphere-Mediated Superhydrophobicity of Rationally Designed Micro/Nanostructured Surfaces.” *ACS Nano*, **13** (4) 4160–4173 (2019)
46. Zhan, D, Guo, Z, “Overview of the Design of Bionic Fine Hierarchical Structures for Fog Collection.” *Materials Horizons*, **10** (11) 4827–4856 (2023)
47. Zhang, L, et al. “Inkjet Printing for Direct Micropatterning of a Superhydrophobic Surface: Toward Biomimetic Fog Harvesting Surfaces.” *Journal of Materials Chemistry A*, **3** (6) 2844–2852 (2015)
48. Ahmed, G, et al. “Modeling the Effects of Contact Angle Hysteresis on the Sliding of Droplets Down Inclined Surfaces.” *European Journal of Mechanics-B/Fluids*, **48** 218–230 (2014)
49. Carlson, A, “Capillarity and Dynamic Wetting.” PhD Thesis, KTH Royal Institute of Technology (2012)
50. Furmidge, C, “Studies at Phase Interfaces. I. The Sliding of Liquid Drops on Solid Surfaces and a Theory for Spray Retention.” *Journal of Colloid Science*, **17** (4) 309–324 (1962)
51. Kim, H-Y, Lee, HJ, Kang, BH, “Sliding of Liquid Drops Down an Inclined Solid Surface.” *Journal of Colloid and Interface Science*, **247** (2) 372–380 (2002)
52. Abosedira, S, et al. “A Practical Strategy of Electrospun Fibers of Polystyrene/Cellulose Acetate Blend for Atmospheric Water Harvesting.” *Alexandria Engineering Journal*, **111** 579–587 (2025)
53. Zhang, Y, et al. “Atmospheric Water Harvesting by Large-Scale Radiative Cooling Cellulose-Based Fabric.” *Nano Letters*, **22** (7) 2618–2626 (2022)